

PROFESSIONAL SUMMARY

Senior AI Engineer at the intersection of computer science and finance. I build **production agentic AI** and work **forward deployed**, ramping up quickly, listening for what clients actually need, and going **full stack** from architecture and **agents** through UI and shipping. I owned an agentic AI **platform** in **production**, from **proof of concept** to years of active use. I maintain **digithings**, an **open-source** AI infrastructure stack, and build tools on it for clients.

PROJECTS

digithings.ai

Open-source AI library and toolkit. It is a canvas for agentic ideas and the layer I use to build production applications. Composable modules for orchestration, RAG/retrieval, DigiChat surfaces, auth, and observability; self-hosted by default (MIT), BYOK, swappable vector backends.

- **DataTap Stream Chat**: DigiChat release embedded on a client’s product documentation, with support chat and developer/docs chat in place and retrieval against the live docs.

digiquant.io

Automated quant finance application built on digithings. Agentic research produces ranked, sourced theses; multi-agent deliberation shapes portfolio direction and risk sizing; then indicators, strategy specs, Optuna search, and reproducible NautilusTrader backtests with trade ledgers and tearsheets. Research book and paper portfolio ship today; live broker execution is not built.

- **FX Hub**: FX research with traders on bank and broker consensus and trade ideas.

EXPERIENCE

Senior AI Engineer

- PSP Investments, Alpha Science, Montreal, QC
- Sep 2022 - Aug 2026
- Owned the full stack of PSP Investments' company-wide agentic AI platform for analysts and PMs: architecture, design documentation, and UI/UX from concept to deployment; feature development, SDK packaging & distribution, PR reviews, and ongoing maintenance. Source-traced answers land in daily research workflows.
 - **Alpha Science SDK**: wraps Databricks (Vector Search, SQL Warehouses, Genie Agents) & OpenAI (Responses + Files API), auto-auth via Databricks CLI & Azure Key Vault. Composable agent/workflow base-class framework on the firm’s existing stack, proprietary tracing, inter-agent persistence, pre-built financial utility agents & skills; artifact export (Office docs & web pages), sub-agent delegation with to-do planning, and parallel tool execution.
 - **Virtual Analyst Platform**: custom React, NestJS, and FastAPI frontend (refactored from Gradio) that became the shell for company AI tools and dashboards. Modern chat UI with a side panel for source citation and artifact viewing; workflow launcher for long-running research tasks; task manager with subscription-style pre-written prompts (e.g. digests). Step-level tracing, source traceability, structured data analysis, and live charting (Plotly/Dash/ECharts).
 - **Document rebuild pipeline**: from messy retrieval-index text, parallel LLM reformatting into Markdown/PDF reconstructions of filings, transcripts, and broker research so users can read full sourced documents in context, with the original text kept as source of truth.
 - **Agile roadmap**: weekly-evolving priorities with rapid integration of emerging model and tooling capabilities into production. Each feature independently scoped, designed as an SDK tool, and surfaced through the platform from concept to deployment.
 - **Leadership**: direct collaboration with C-suite across all asset classes; mentored interns and juniors to ship production features independently; coached Alpha Science SDK users; platform demoed at the Databricks Data & AI Summit.

Analyst

- PSP Investments, Digital Innovation, Montreal, QC
- Sep 2021 - Aug 2022
- Co-developed **Earnings Prediction Engine** (ML beat/miss pipeline, point-in-time data, Power BI); built alt-data pipeline with data profiling, cleaning & feature engineering for equity factor research; delivered POC analyses to drive stakeholder buy-in across asset class teams

Intern

- PSP Investments, Information Security, Montreal, QC
- May 2021 - Aug 2021
- Enterprise risk framework implementation; automated risk assessment repository updates, improving data currency and reporting efficiency

Business Analyst

- Lunch à Porter, Montreal, QC
- Jun 2020 - Jun 2021
- SEO & A/B testing drove 2x online conversion rate; inventory analysis for data-informed merchandising decisions

SKILLS

- Gen AI & LLMs:** LangGraph, LangChain, LiteLLM, RAG, OpenAI API, Gemini API, Claude API, MCP, Gradio, Azure AI Search, Azure AI Foundry, Prompt Engineering, Vector DBs
- AI Dev Tools:** Cursor, GitHub Copilot, Claude Code, Codex, OpenCode, LM Studio, Open WebUI, Ollama
- Data & Cloud:** Azure, GCP, Databricks, Spark, Delta Lake, PostgreSQL, SQL, Supabase, Docker, MLOps, CI/CD
- Programming:** Python, TypeScript, JavaScript, React, NestJS, Next.js, HTML, CSS, FastAPI, Pydantic, pandas, NumPy, REST APIs, SDK Design, Git
- Visualization:** Power BI, Plotly, Dash, Matplotlib, ECharts, Tableau
- Markets & Data:** QuantConnect, IBKR API, Alpaca API, LSEG, SEC Filings, Broker Research, Earnings Transcripts, Third Bridge

EDUCATION

- McGill University** · B.Com Finance & Business Analytics · Desautels · Montreal, QC · 2019 - 2022
- Champlain College** · DEC Computer Science & Mathematics · Montreal, QC · 2017 - 2019 · Honor Roll

CERTIFICATIONS

- Azure AI Fundamentals** · Microsoft · Sep 2023

LANGUAGES

- English (Native), French (Native), Romanian (Native), Spanish (Basic), Italian (Basic)